Guojie Zhong

PhD in Systems Biology, Columbia University
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RESEARCH INTEREST

My research centers on one question: **How does genetic variation cause disease?** I develop AI and statistical methods to address this question at several levels:

- **Variant effects:** How do variants change protein function and gene regulation? (*PreMode*, *EpiBRAIN*)
- **Genes and pathways:** Which genes and pathways are disrupted in disease? (*VBASS*, *EA/TEF study*)
- **Cellular context:** In which cell types and developmental stages do genetic changes matter? (*VBASS*, *EpiBRAIN*).
- **Tissue organization:** How do cells interact and organize within tissues, and how are these relationships altered in disease? (*CSOmap*)

EDUCATION

08/2019 – 11/2024 PhD in Systems Biology, Integrated program in cellular, molecular and biomedical sciences
Department of Systems Biology, **Columbia University**, New York, NY, USA.
Thesis: Computational approaches to understand mechanisms of human genetic disorders
Thesis Advisor: Yufeng Shen, PhD
Committee: Wendy Chung, MD, PhD; Mohammad AlQurashi, PhD; David Knowles, PhD

08/2017 – 12/2017 Exchange Student in Berkeley Bioscience Study Abroad Program (BBSA)
Department of Integrated Biology, **University of California, Berkeley**, CA, USA.

09/2015 – 07/2019 Bachelor of Science, Integrated Science Program
Yuanpei College, **Peking University**, Beijing, China.
Thesis: Reconstruction of cell spatial organization from single-cell RNA sequencing data based on ligand-receptor mediated self-assembly
Thesis Advisor: Zemin Zhang, PhD

PUBLICATIONS

(# = co-first author)

Deep Learning for gene regulation and non-coding variant effect prediction

- Chang, L.#, Li, K.#, Xie, Y.#, **Zhong, G.**#, Rink, J.A., Baez-Becerra, C.T., Lie, A., Indralingam, H.S., Dong, K., Loe, T., et al. Single-cell Multiome Analysis of Chromatin State and Transcriptome in the Human Basal Ganglia. *Cell (in press)*. 10.64898/2026.02.03.703645 (2026).
- Xie, Y.#, Chang, L.#, **Zhong, G.**, Rink, J.A., Baez-Becerra, T., Armand, E., Ding, W., Li, K., Bonne, E., Lie, A., et al. Single-Cell Atlas of Transcription and Chromatin States Reveals Regulatory Programs in the Human Brain. *Cell (in press)*. 10.64898/2026.02.02.703166 (2026).

Deep Learning for coding variant effect prediction

- **Zhong, G.**, Zhao, Y., Zhuang, D., Chung, W. K. & Shen, Y. PreMode predicts mode-of-action of missense variants by deep graph representation learning of protein sequence and structural context. *Nat Commun* **16**, 7189 (2025).
- Zhao, Y., Lan, T., **Zhong, G.**, Hagen, J., Pan, H., Chung, W.K., and Shen, Y. A probabilistic graphical model for estimating selection coefficients of nonsynonymous variants from human population sequence data. *Nat Commun* **16**, 4670 (2025).

Statistical genetics and rare diseases

- **Zhong, G.**, Choi, Y.A. and Shen, Y. VBASS enables integration of single cell gene expression data in Bayesian association analysis of rare variants. *Commun Biol* **6**, 774 (2023).
- **Zhong, G.**, and Shen, Y. Statistical models of the genetic etiology of congenital heart disease. *Curr Opin Genet Dev* **76**, 101967 (2022).
- **Zhong, G.**#, Ahimaz, P.#, Edwards, N.A.#, Hagen, J.J., Faure, C., Lu, Q., Kingma, P., Middlesworth, W., Khlevner, J., El Fiky, M., et al. Identification and validation of candidate risk genes in endocytic vesicular trafficking associated with esophageal atresia and tracheoesophageal fistulas. *HGG Adv* **3**, 100107 (2022).
- Edwards, N., **Zhong, G.**, Ahimaz, P., Kenny, A., Kingma, P., Wells, J., Shen, Y., Chung, W., and Zorn, A. Discovering the Developmental Basis of Trachea-Esophageal Birth Defects: Evidence for Endosome-opathies. *The FASEB Journal* **36** (2022).

Single cell and spatial transcriptomics

- Su, J., Reynier, J.B., Fu, X., **Zhong, G.**, Jiang, J., Escalante, R.S., Wang, Y., Aparicio, L., Izar, B., Knowles, D.A., and Rabadan, R. Smoother: a unified and modular framework for incorporating structural dependency in

spatial omics data. Genome Biol **24**, 291(2023).

- Ren, X.#, **Zhong, G.**#, Zhang, Q., Zhang, L., Sun, Y., and Zhang, Z. Reconstruction of cell spatial organization from single-cell RNA sequencing data based on ligand-receptor mediated self-assembly. *Cell Res* **30**, 763-778 (2020).
- Zhang, Q.#, He, Y.#, Luo, N.#, Patel, S.J., Han, Y., Gao, R., Modak, M., Carotta, S., Haslinger, C., Kind, D., Peet G.W., **Zhong, G.**, Lu, S., Zhu, W., Mao, Y., Xiao, M., et al. Landscape and Dynamics of Single Immune Cells in Hepatocellular Carcinoma. *Cell* **179**, 829-845 e20 (2019).

CONFERENCES

- **Zhong, G.**, and Shen, Y. (2023). Predicting mode of action of missense variants by graph representation of protein structural context. *American Society of Human Genetics 2023 Annual Meeting,* Washington, D.C.
- **Zhong, G.**, and Shen, Y. (2022). Representation of missense variants for predicting modes of action. *Machine Learning in Structural Biology, Workshop at the 36th Conference on Neural Information Processing Systems,* New Orleans, LA.
- **Zhong, G.**, Choi, Y.A., and Shen, Y. (2022). Integration of gene expression data in Bayesian association analysis of rare variants. *American Society of Human Genetics 2022 Annual Meeting,* Los Angelas, CA.
- **Zhong, G.**, Wang, J., He, S. and Fu, X. (2021). Towards better understanding of developmental disorders from integration of spatial single-cell transcriptomics and epigenomics. *The 2021 ICML Workshop on Computational Biology,* Virtual meeting.

AWARDS

2024	Chinese Government Award for Outstanding Self-Financed Students Abroad
2019	Dean's fellowship, Graduate School of Arts and Science, Columbia University
2015	2014-2015 academic year Outstanding Freshman Scholarship, Peking University
2014	The 28th National Olympiad in Chemistry in Provinces, 1st prize, Zhejiang Province, China

OPEN-SOURCE SOFTWARES

2025-2026	EpiBRAIN	https://github.com/zhongguojie1998/EpiBRAIN Sequence-to-function model for predicting cell-type-specific chromatin profiles and gene expression from DNA sequence.
2022-2024	PreMode	https://github.com/ShenLab/PreMode Pretrained SE(3)-equivariant graph neural network model for missense variant effect representation that can accurately predict mode-of-action of missense variants using protein language models and structural context.
2020-2022	VBASS	https://github.com/ShenLab/VBASS Semi-supervised machine learning algorithm that combines neural network with Bayesian statistical models to identify disease risk genes for rare genetic conditions jointly from scRNA-seq data and rare variants data.
2018-2020	CSOmap	https://github.com/zhongguojie1998/CSOmap Unsupervised machine learning algorithm to reconstruct single cell spatial information and infer cellular interactions in tumor microenvironments from scRNA-seq data.

MENTORSHIP EXPERIENCES

05/2025 – 08/2025
Di Liu, summer rotation PhD student
Ren Lab, New York Genome Center
101 6th Ave, New York, NY, USA

06/2023 – 08/2023
06/2022 – 08/2022
Demi Zhuang, summer visiting undergraduate student
Shen Lab, Department of Systems Biology
Columbia University Irving Medical Center, New York, NY, USA
Developed protein family specific benchmarking tasks and datasets for PreMode. Applied PreMode to those datasets and discovered several novel likely G/LoF variants.

06/2021 – 08/2021
Andrew Lee, summer visiting high school student
Shen Lab, Department of Systems Biology
Columbia University Irving Medical Center, New York, NY, USA

Developed an R app to visualize the scRNA-seq data of autism risk genes in brain cell types from public datasets. Applied it to discovery of novel autism risk genes and identification of disease associated cell types.

PROFESSIONAL EXPERIENCES

07/2026 – present	Genomics AI fellow
03/2025 – 06/2026	Postdoctoral Research Associate Ren Lab, New York Genome Center 101 6th Ave, New York, NY, USA
01/2025 – 02/2025	Postdoctoral Research Scientist
09/2019 – 12/2024	Graduate Research Assistant Shen Lab, Department of Systems Biology Columbia University Irving Medical Center, New York, NY, USA
06/2024 – 09/2024	Data Science & Machine Learning Intern Insitro, Inc. South San Francisco, CA, USA
03/2017 – 07/2019	Research Assistant Zhang Lab, Biomedical Pioneering Innovation Center (BIOPIC) Peking University, Beijing, China

ACADEMIC AND VOLUNTEER SERVICE

Peer Review Service

PLOS Computational Biology; Molecular Medicine; Scientific Reports; PeerJ; Machine Learning in Structural Biology Workshop (NeurIPS 2022).

REFERENCES

Bing Ren, Ph.D.,
Professor, Department of Genetics and Development, Columbia University Irving Medical Center
Professor, Department of Biochemistry and Molecular Biophysics, Columbia University Irving Medical Center
Professor, Department of Systems Biology, Columbia University Irving Medical Center
Associate Director, the Roy and Diana Vagelos Institute for Basic Biomedical Science
Scientific Director and Chief Executive Officer, New York Genome Center
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Relationship: Postdoc Advisor

Yufeng Shen, Ph.D.,
Associate Professor, Department of Systems Biology, Columbia University Irving Medical Center
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Relationship: PhD Advisor

Wendy K. Chung, M.D., Ph.D.,
Chief, Department of Pediatrics, Boston Children's Hospital
Mary Ellen Avery Professor of Pediatrics, Harvard Medical School
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Assistant Professor, Department of Systems Biology, Columbia University Irving Medical Center
Assistant Professor, Department of Computer Science, Columbia University
Email: ma4129@cumc.columbia.edu
Relationship: PhD Thesis Committee Member

David A. Knowles, Ph.D.,
Associate Professor, Department of Computer Science, Columbia University
Associate Professor, Department of Systems Biology, Columbia University Irving Medical Center
Affiliate Member, Data Science Institute, Columbia University
Core Faculty Member, New York Genome Center
Email: dak2173@columbia.edu
Relationship: PhD Thesis Committee Member